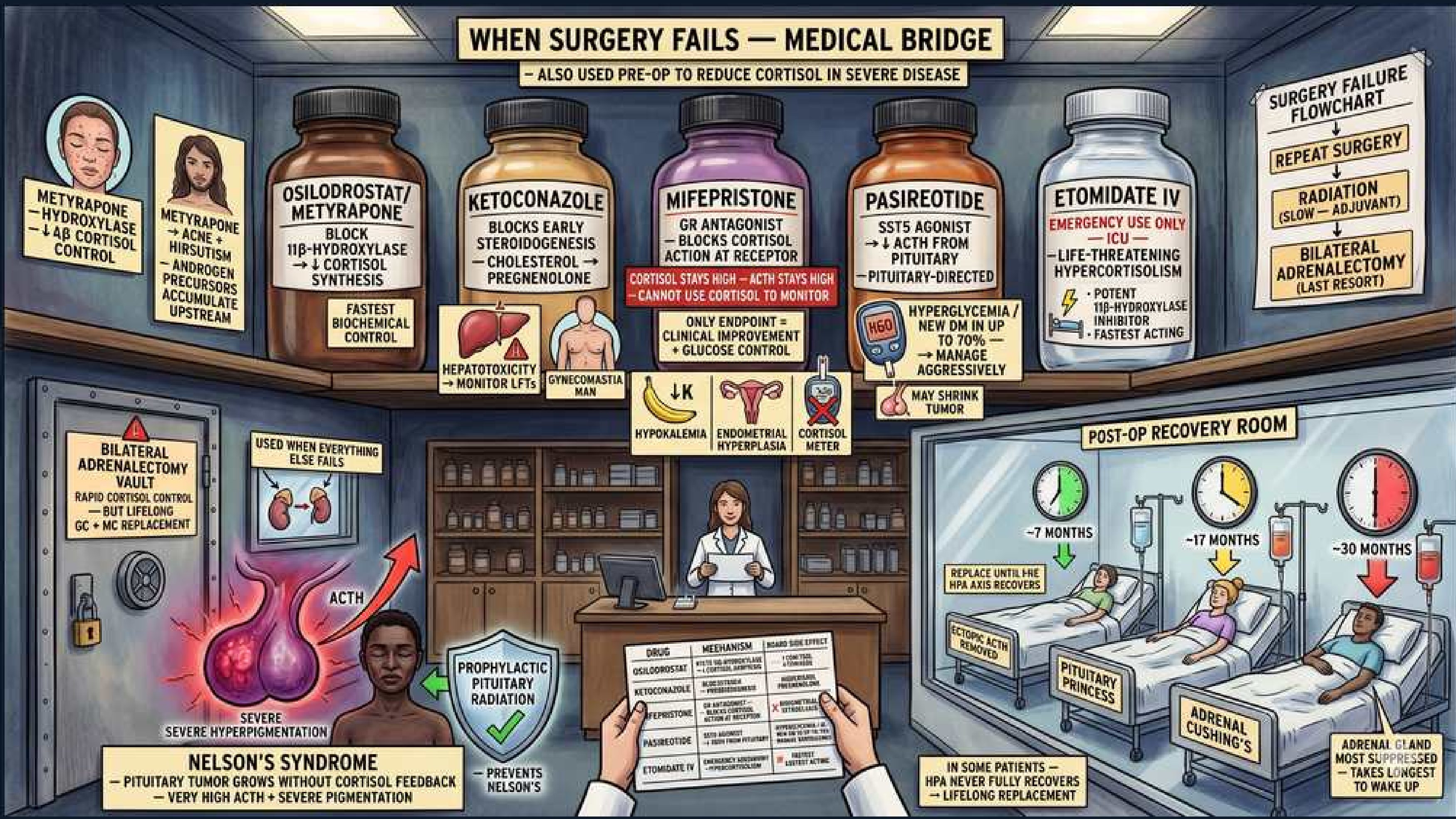
Adrenal — Cushing Treatment Pharmacy



1. Surgery is first-line

WHAT YOU SEE

Banner 'WHEN SURGERY FAILS — MEDICAL BRIDGE'

WHAT IT ENCODES

Definitive cure for endogenous Cushing is SURGERY: transsphenoidal selective adenomectomy for Cushing DISEASE (pituitary ACTH adenoma, ~80% of ACTH-dependent cases), unilateral adrenalectomy for a cortisol-secreting adrenal adenoma/ACC, and resection of the ectopic source for ectopic ACTH (e.g. bronchial carcinoid, small-cell lung). Drugs are a BRIDGE — used pre-op to lower cortisol in severe disease, while awaiting radiation effect, or when surgery fails/is not feasible. Remission after transsphenoidal surgery is judged by a low/undetectable post-op morning cortisol.

BOARD TRAP

Classic trap: choosing medical therapy as the definitive 'treatment of choice' — it is adjunctive/bridging, never the curative first step when the source is resectable.

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2. Ketoconazole

WHAT YOU SEE

Bottle 'KETOCONAZOLE — blocks steroidogenesis, cholesterol→pregnenolone'

WHAT IT ENCODES

Ketoconazole is an azole antifungal repurposed as a steroidogenesis inhibitor: it blocks several CYP enzymes (side-chain cleavage cholesterol→pregnenolone, 11β-hydroxylase, 17α-hydroxylase). Key toxicities are HEPATOTOXICITY (monitor LFTs, can be fatal — black-box warning) and, because it also blocks gonadal androgen synthesis, GYNECOMASTIA and decreased libido in men. Needs gastric acid for absorption, so avoid PPIs.

BOARD TRAP

Don't co-prescribe with a PPI/H2-blocker (impairs absorption) and don't ignore rising transaminases — hepatotoxicity is the board-tested adverse effect.

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3. Metyrapone

WHAT YOU SEE

Bottle 'METYRAPONE — 11β-hydroxylase block, fastest biochemical control'

WHAT IT ENCODES

Metyrapone inhibits 11β-hydroxylase, the FINAL step (11-deoxycortisol→cortisol), giving fast biochemical control of hypercortisolism. Because the block is downstream, precursors (11-deoxycortisol) and ADRENAL ANDROGENS accumulate → acne and HIRSUTISM in women, plus the mineralocorticoid precursor 11-deoxycorticosterone (DOC) can cause HYPERTENSION/HYPOKALEMIA/edema.

BOARD TRAP

Wrong to assume metyrapone is androgen-neutral — it worsens hirsutism/acne and can cause mineralocorticoid-excess HTN, unlike receptor-blocker mifepristone.

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4. Osilodrostat

WHAT YOU SEE

Bottle 'OSILODROSTAT — 11β-hydroxylase, cortisol synthesis block'

WHAT IT ENCODES

Osilodrostat is a potent oral 11β-hydroxylase inhibitor (mechanistically a metyrapone successor) approved specifically for Cushing disease. It produces strong, sustained cortisol normalization but, like metyrapone, shunts precursors upstream → accumulation of androgens (hirsutism) and the mineralocorticoid DOC → hypokalemia, edema, HTN; also QT prolongation.

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5. Mifepristone

WHAT YOU SEE

Bottle 'MIFEPRISTONE — GR antagonist, blocks cortisol at receptor'

WHAT IT ENCODES

Mifepristone is a glucocorticoid-RECEPTOR antagonist (also a progesterone antagonist), used for Cushing-associated HYPERGLYCEMIA/diabetes when surgery fails. Because it blocks the receptor rather than synthesis, serum/urine cortisol and ACTH actually RISE — you must follow CLINICAL/metabolic endpoints (glucose, weight, BP), not cortisol levels. Excess unopposed cortisol at the mineralocorticoid receptor causes HYPOKALEMIA and HTN; in women it causes endometrial hyperplasia/vaginal bleeding.

BOARD TRAP

High-yield trap: you CANNOT monitor mifepristone with cortisol levels (they go up by design) — monitor glucose/clinical response. Also watch for adrenal-insufficiency symptoms with normal/high labs.

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6. Pasireotide

WHAT YOU SEE

Bottle 'PASIREOTIDE — SSTR, lowers ACTH from pituitary'

WHAT IT ENCODES

Pasireotide is a multireceptor somatostatin analog (high affinity for SSTR5, expressed on corticotroph adenomas) that lowers pituitary ACTH secretion and can shrink the tumor — it targets the SOURCE in Cushing disease. Its signature adverse effect is HYPERGLYCEMIA/new diabetes (suppresses insulin and incretins) in a majority of patients; also gallstones and bradycardia.

BOARD TRAP

Don't forget hyperglycemia — paradoxically the cortisol-lowering pasireotide commonly causes/worsens diabetes, so glucose must be monitored.

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7. Etomidate IV (emergency)

WHAT YOU SEE

Bottle 'ETOMIDATE IV — emergency life-threatening hypercortisolism, fastest acting'

WHAT IT ENCODES

Etomidate is an IV anesthetic that potently inhibits 11β-hydroxylase, making it the FASTEST agent to drop cortisol — reserved for acute, life-threatening hypercortisolemic crisis (severe psychosis, sepsis, uncontrolled hyperglycemia/hypertension) when oral therapy isn't feasible. Given at sub-hypnotic doses with ICU monitoring; may require concurrent hydrocortisone 'block-and-replace.'

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8. Bilateral adrenalectomy vault

WHAT YOU SEE

Vault 'BILATERAL ADRENALECTOMY — when everything else fails, lifelong GC+MC replacement'

WHAT IT ENCODES

Bilateral adrenalectomy is the LAST-RESORT definitive option (failed pituitary surgery/radiation, severe ectopic or occult ectopic ACTH) — it immediately and reliably abolishes hypercortisolism. The cost is permanent primary adrenal insufficiency requiring LIFELONG glucocorticoid AND mineralocorticoid (fludrocortisone) replacement, plus the risk of Nelson syndrome.

BOARD TRAP

Trap: forgetting that mineralocorticoid replacement (fludrocortisone) is also required — unlike pituitary-cause hypoadrenalism, bilateral adrenalectomy removes aldosterone production too.

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9. Nelson syndrome

WHAT YOU SEE

Red pituitary tumor + dark-skinned figure 'NELSON SYNDROME — very high ACTH, severe pigmentation'

WHAT IT ENCODES

Nelson syndrome occurs after bilateral adrenalectomy for Cushing DISEASE: loss of cortisol negative feedback lets the residual corticotroph adenoma grow aggressively → very high ACTH (and co-secreted POMC/MSH) causing marked HYPERPIGMENTATION, plus mass effect (headache, visual field/chiasm compression). Monitor with serial ACTH and pituitary MRI.

BOARD TRAP

Hyperpigmentation after adrenalectomy = high ACTH (Nelson), not adrenal insufficiency per se — and it occurs only in pituitary-driven (Cushing disease) cases, not adrenal/ectopic sources.

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10. Prophylactic pituitary radiation

WHAT YOU SEE

'PROPHYLACTIC PITUITARY RADIATION — prevents Nelson'

WHAT IT ENCODES

Prophylactic pituitary radiation (or radiosurgery) before/at the time of bilateral adrenalectomy reduces the risk of subsequent corticotroph tumor enlargement (Nelson syndrome). Radiation is also a slow-acting second-line option for Cushing disease itself, with effect taking months to years and a risk of subsequent hypopituitarism.

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11. Mitotane for ACC

WHAT YOU SEE

Skull bottle 'MITOTANE — adrenolytic in ACC'

WHAT IT ENCODES

Mitotane (o,p'-DDD) is an ADRENOLYTIC/adrenocorticolytic agent (DDT derivative) that destroys adrenocortical cells and inhibits steroidogenesis — the cornerstone medical therapy for ADRENOCORTICAL CARCINOMA (adjuvant after resection and in advanced disease), not for benign Cushing. It is slow and toxic: GI upset, neurotoxicity (ataxia/confusion), and it INDUCES CYP3A4 and raises CBG, so it causes adrenal insufficiency requiring higher-than-usual hydrocortisone replacement. Narrow therapeutic window (target ~14–20 mg/L).

BOARD TRAP

Don't pick mitotane for ordinary Cushing disease/adenoma — it's the ACC drug. Also remember it induces metabolism and causes adrenal insufficiency, so steroid replacement doses must be increased.

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12. Post-op recovery / HPA recovery

WHAT YOU SEE

Beds '~7 / ~17 / ~30 months', adrenal insufficiency, lifelong replacement

WHAT IT ENCODES

After curative removal of a cortisol source, the chronically suppressed contralateral adrenal/HPA axis cannot immediately resume cortisol output → transient (sometimes prolonged) secondary adrenal insufficiency. Patients need glucocorticoid replacement (hydrocortisone) with a slow taper guided by morning cortisol/ACTH-stimulation testing; recovery typically takes MONTHS (the beds depict ~7, ~17, ~30 months), occasionally longer. Counsel on steroid sick-day rules and crisis prevention.

BOARD TRAP

Trap: stopping glucocorticoid replacement abruptly post-op — HPA recovery takes many months and abrupt withdrawal precipitates adrenal crisis. Watch for steroid-withdrawal symptoms even with 'normalized' cortisol.
